

CS/EE 120B Custom Laboratory Project Report

Project Title: Space Shooter

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Introduction

The project was successfully implemented with 3 complexities that are as follows: Use of 128x128 LCD, complex game logic, and use of 16x2 LCD with custom characters. The game can be started and stopped at any time using two buttons. Character can be moved in 4 directions on the screen. 8 bullets will be shot out from the character automatically in the 8 directions with a straight pathway. The collision of a bullet and a mob causes the mob and the bullet to despawn, with a single point incrementation of score. The collision between a mob and the character will decrease HP (health points) by 1, and when all 5 HPs are lost, the game is over. The HP and current score can be seen on the 16x2 LCD during gameplay. The 128x128 LCD will display the final score when the game is over.

Build Upons

- 128x128 LCD
 - Fully implemented without errors
- 16x2 LCD with custom character
 - Fully implemented without errors
- Complex game logic
 - Fully implemented without errors

User Guide

Press L_Button (start) to start the game.

Use the joystick to move the character.

Try to dodge mobs and score points.

Press R_Button (reset) during gameplay to stop the game.

The game can be restarted by pressing the L_Button again.

Hardware Components

Computing:

- Elegoo UNO R3.

Inputs:

- Joystick
- Reset, and Start Button.

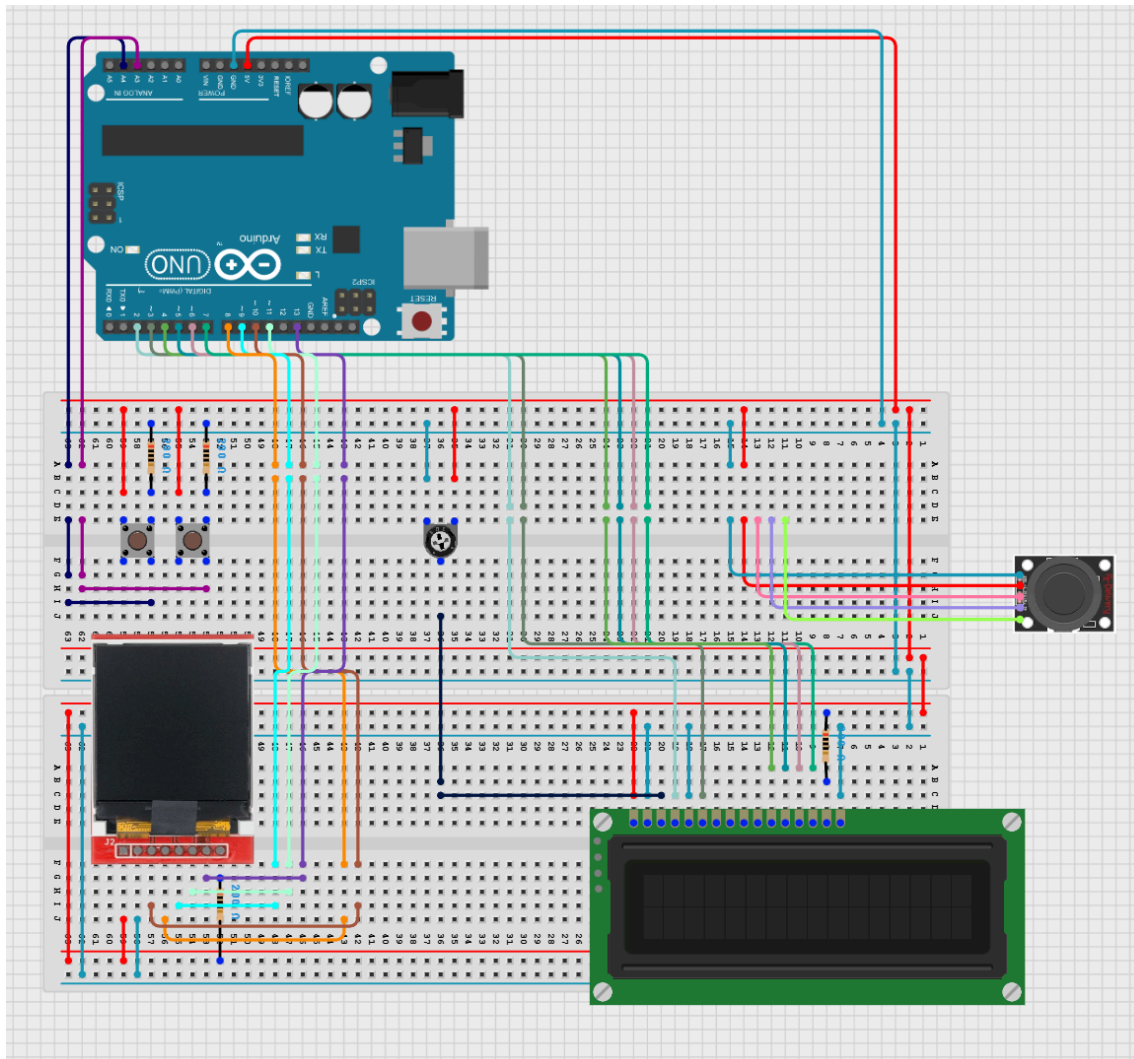
Outputs:

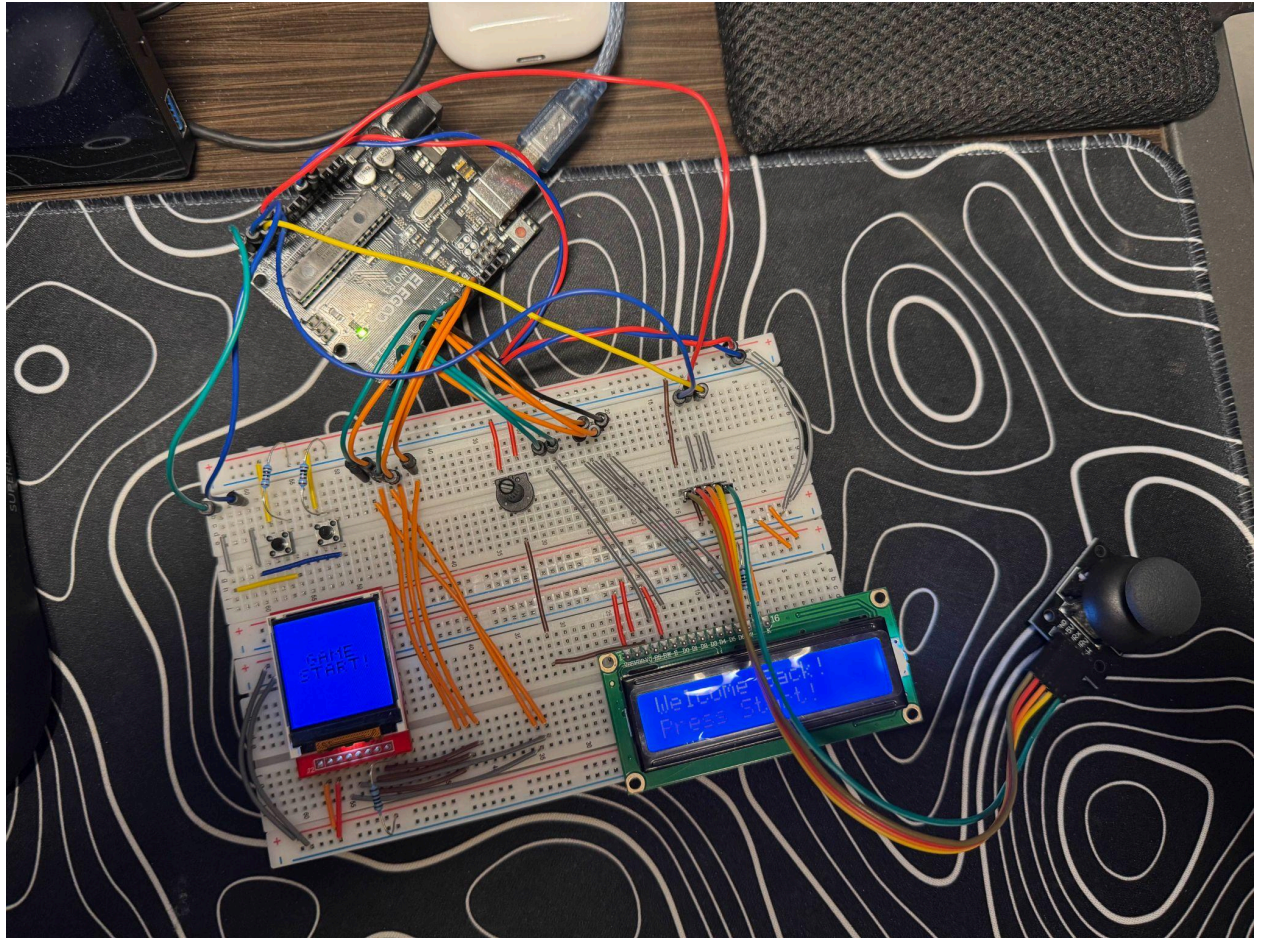
- HiLetgo HT7735 LCD Screen
 - Used to display the game
- 16x2 LCD

External Software Libraries Used

- Time.h
 - Used to generate a random number to spawn mobs randomly on the screen.

Wiring Diagram



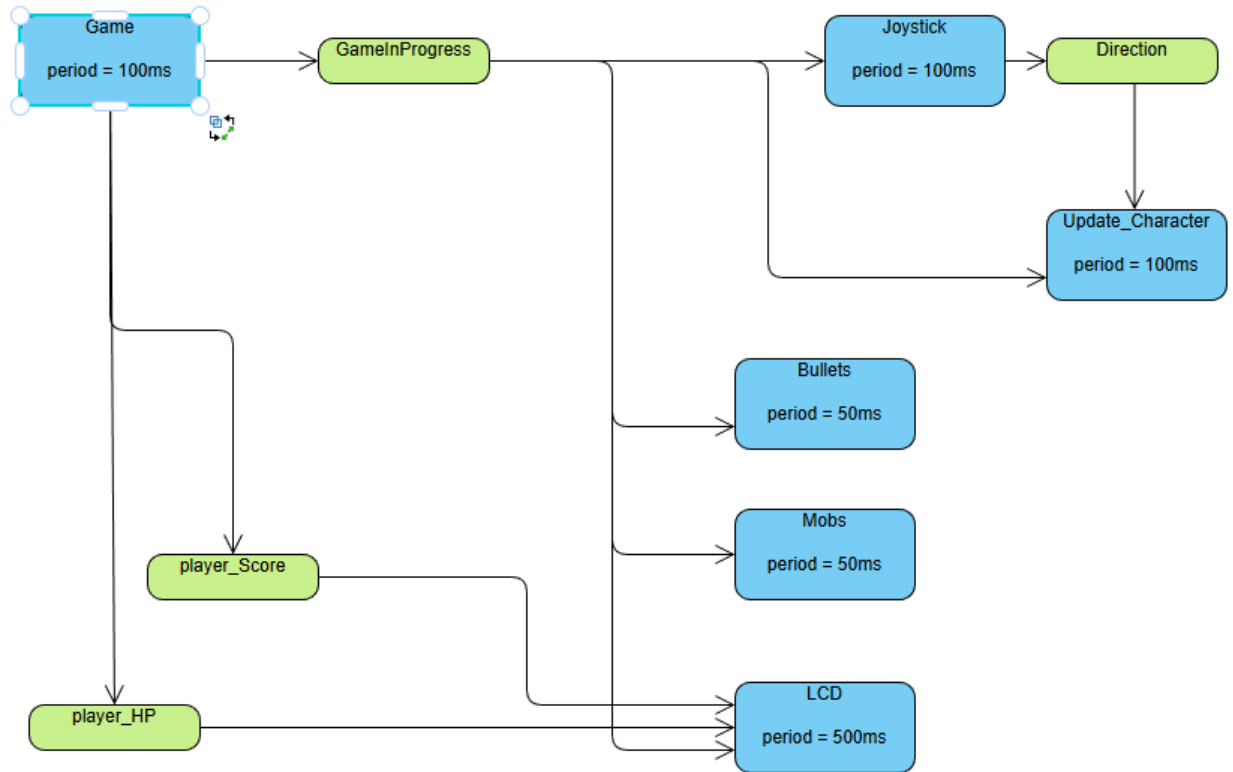


Task Diagram

Period: 50ms

unsigned char Direction = 0b0000, GameInProgress = 0, player_Score = 0;

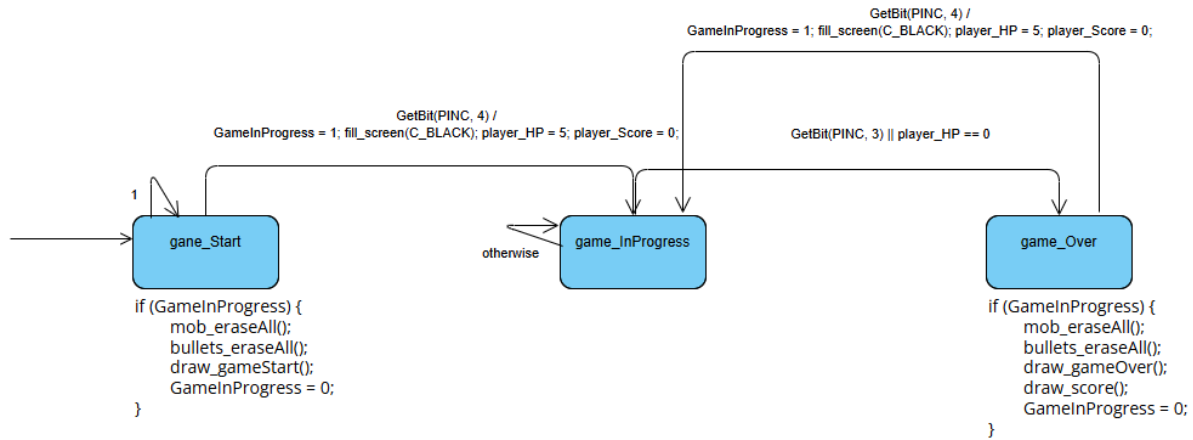
char player_HP = 5;



SynchSM Diagram

Game

Period: 100ms

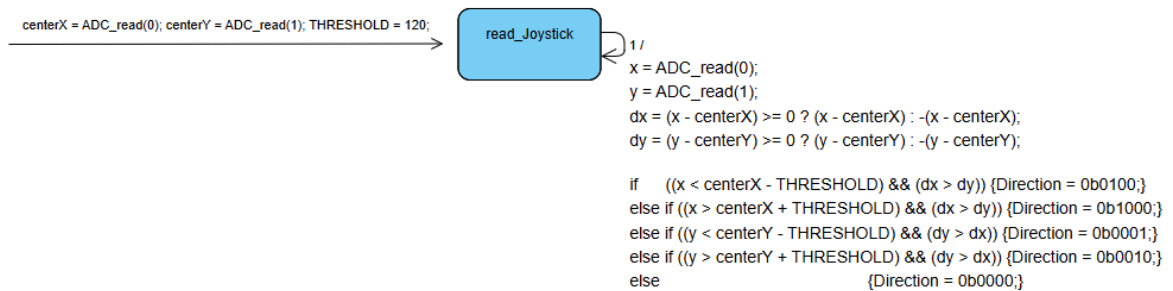


Joystick

Period: 100ms

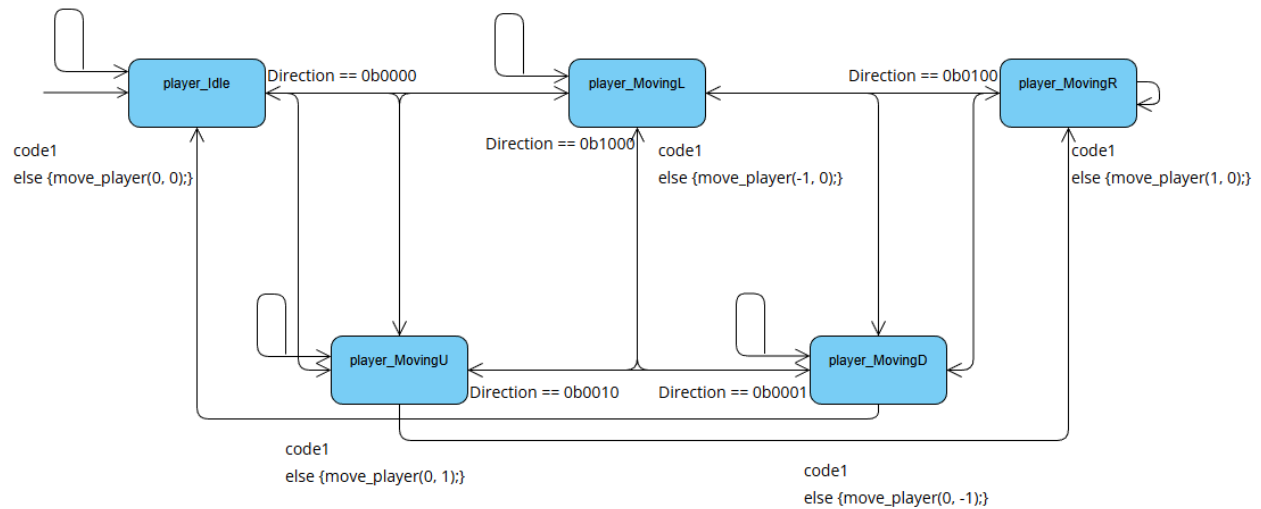
static const int centerX; static const int centerY; const int THRESHOLD;

int x, y, dx, dy;

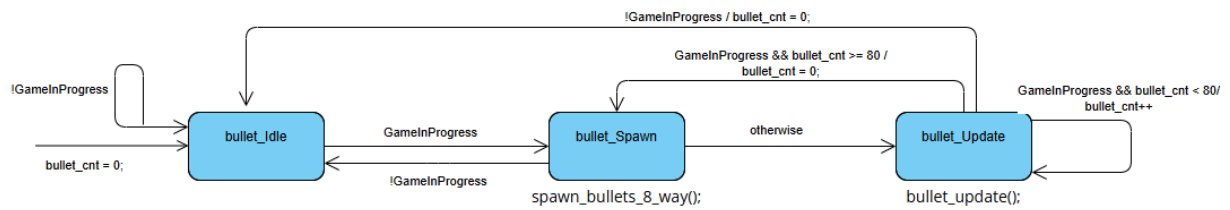


Update_Character
Period: 100ms

Code1:
if (!GameInProgress) {
 player_x = 60;
 player_y = 60;
}



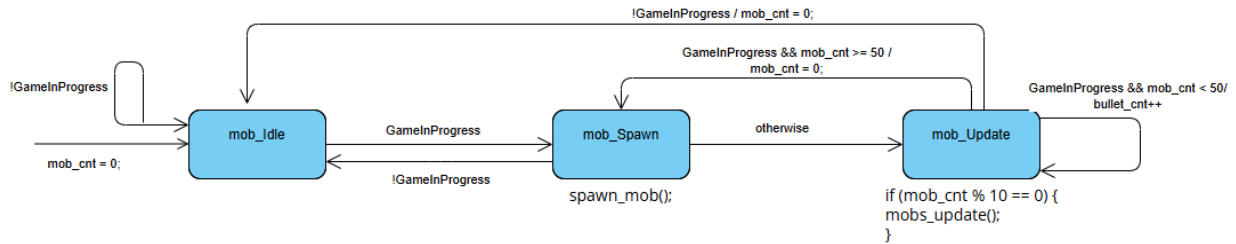
Bullets
Period: 50ms
static unsigned char bullet_cnt;



Mobs

Period: 50ms

static unsigned char mob_cnt;



LCD

Period: 500ms

unsigned char HP_whitespace = 13 - player_HP; char scoreStr[6]; char idleMsg1[17] = "Welcome Back! "; char idleMsg2[17] = "Press Start! ";

char HPMsg[4] = "HP:"; char space[2] = " "; char ScoreMsg[12] = "Score: ";

